Lab Report 1: Magnetoelastic Generators as Self-Powered Bioelectronics	
Bioengr 167: Bioengineering Laboratory 2021 Fall	
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Time and Location	December 3, 2021 12:00 PM Boelter Hall 7736
Abstract	In our lab sections, our team fabricated an MEG-based respiration strain sensor. We fabricated this device using PDMS, 25 µm magnetic nanoparticles, copper wire, aluminum foil, physiotherapy bands, and tape. This device was used to collect normal breathing and hard exhale data by placing the device on the diaphragm and the chest. We measured the current of normal breathing, rapid breathing, and forced expiration for two individuals, and aim to use this data to measure lung tidal volume and forced vital capacity. With this device, we hope to establish a method of timely and correct diagnosis of chronic respiratory diseases and engage in proper symptom management in order to decrease symptoms, reduce the frequency and severity of exacerbations, and improve health status.

Introduction	Chronic respiratory diseases including chronic obstructive pulmonary disease (COPD) and asthma affect more than 435 million people worldwide as of 2019. In the US, COPD is the fourth highest cause of death resulting in approximately 120,000 deaths per year. Establishing a correct diagnosis and proper monitoring of COPD is important because appropriate management can decrease symptoms, especially dyspnea, reduce the frequency and severity of exacerbations, improve health status, improve exercise capacity, and prolong survival. Some ways that physicians currently evaluate respiratory function are through measurements of lung volumes and breathing cycles. The two main measurements when it comes to the diagnosis of COPD is forced expiratory volume after 1 second known as FEV1 and the total volume that is forcibly expired known as the forced vital capacity. As can be seen in Figure 2B the FEV1 of individuals with and without COPD differ quite drastically. The ratio of these values can be a crucial factor in the proper diagnosis of an obstructive disease or a restrictive disease. Analysis of the FEV1 and forced vital capacity can yield a determination of the presence or absence of COPD. Moreover, the configuration of a flow-volume loop can help distinguish the site of airway narrowing. As seen in Figure 2C, the first loop indicates a normal flow-volume loop, the second indicates a dynamic extrathoracic obstruction, the third indicates a dynamic intrathoracic obstruction, the fourth indicates a fixed upper airway obstruction, and the fifth indicates a peripheral or lower airway obstruction. By having continuous monitoring and measurement of these values, disease can be better assessed by healthcare officials and patient health outcomes can improve. Currently, physicians evaluate FEV1 and forced vital capacity through clinical assessments, such as spirometry, full body plethysmography, and arterial blood sampling and diffusion capacity measurements. Because these methods require instrumentation and the aid of a lab technician, they provide an expensive and episodic form of disease diagnosis and treatment. A non-clinical solution to monitor these values exists in the form of a wearable strain sensor, consisting of a piezoresistive metal thin film set in a silicone elastomer substrate. Though providing the accessibility and continuous monitoring that clinical solutions lack, this piezoresistive strain sensor also has its limitations regarding data collection and accuracy. More specifically, the sensor is only sensitive to linear strain, introducing constraints regarding placement and measurement. In addition, researchers found that the flexibility of the silicone elastomer caused the sensor to distort from local deformations on the surface of the skin, providing another source of inaccuracy in the

	device's function. Our goal is to combat these limitations by shifting the current strain sensor's reliance on piezoresistivity to the giant magnetoelastic effect; with the magnetoelastic generator (MEG), we've designed a self-powered, wearable strain sensor for continuous-monitoring of respiratory health and function. The proposed use of an MEG sensor for this application is promising as it is intrinsically waterproof, yielding accurate results even if the user is sweating during strenuous exercise. Additionally, the need to simply measure the strain associated with the expansion of the user's chest as a function of the force exhaled volume is optimal for utilization of the MEG as it has been proven to be an accurate self-powered strain sensor. The low mechanical modulus of the MEG sensor makes it conformable to the movement and curves of the human body, and its thin nature makes it a sleek alternative to the current rather bulky solutions. Additionally, as can be seen in Figure 1 there are a lot of muscles involved in the action of breathing, all of which we can monitor with this proposed MEG solution. Since we want accurate measurements on the force exhaled volume within the first second of exhalation and the total force exhaled volume, we propose a film MEG sensor applied to the chest of the user rather than a textile MEG, which has been used to provide respiratory information on breathing types (normal, rapid, coughing) but likely will not be able to yield accurate amplitude data for the application we are targeting.

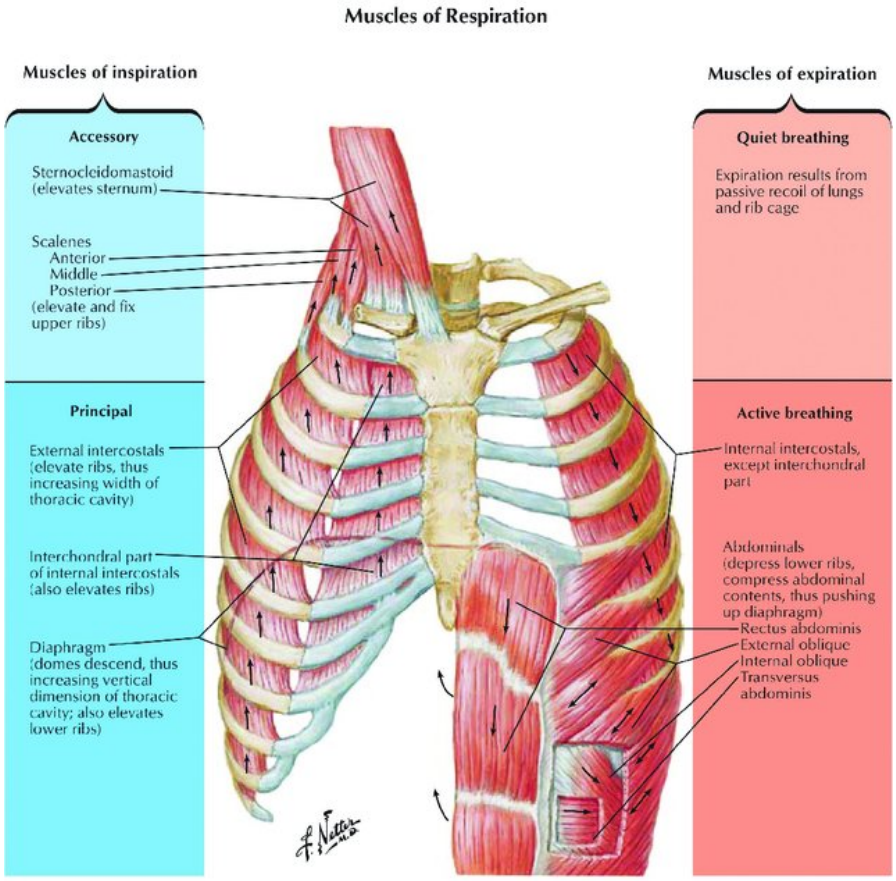
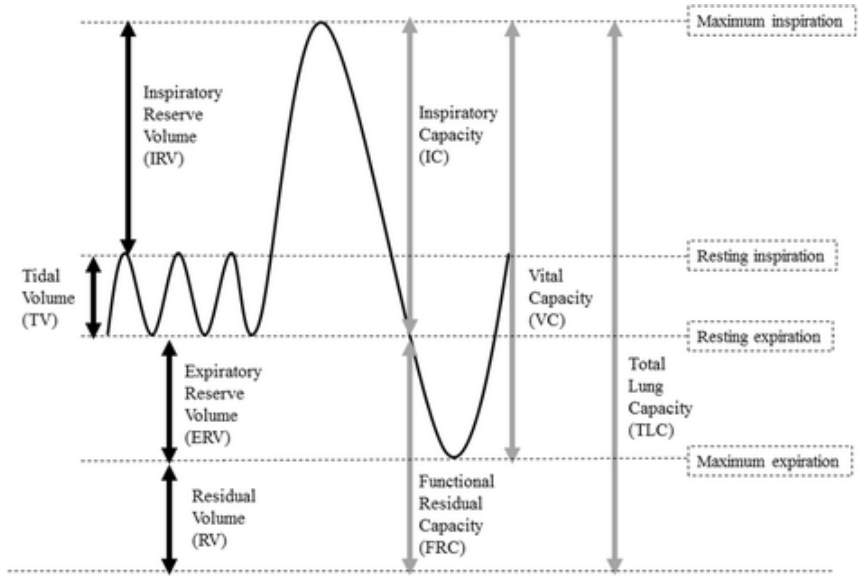


Figure 1. Muscles of Respiration Courtesy of Netter et al, Physiology 1989.

A.



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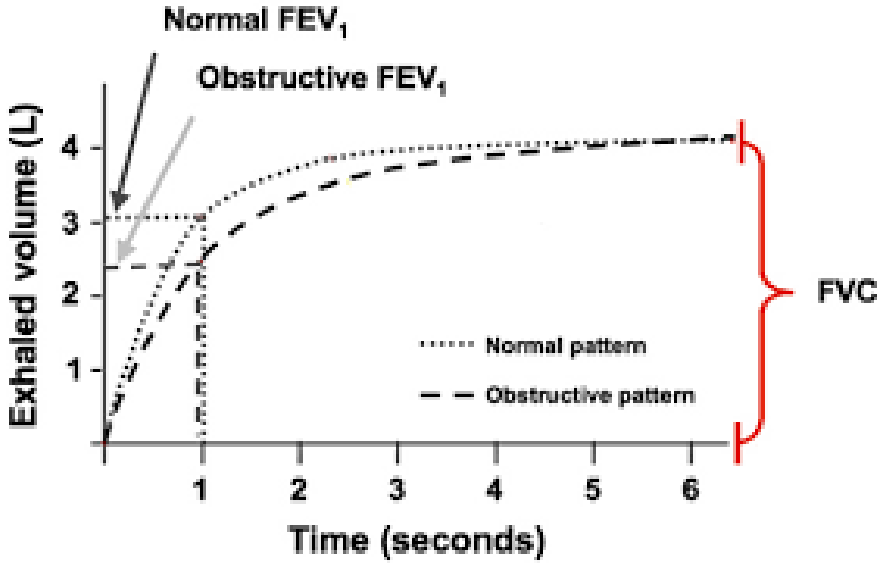

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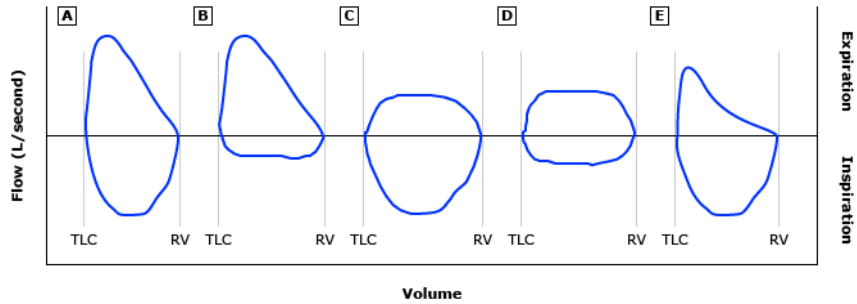
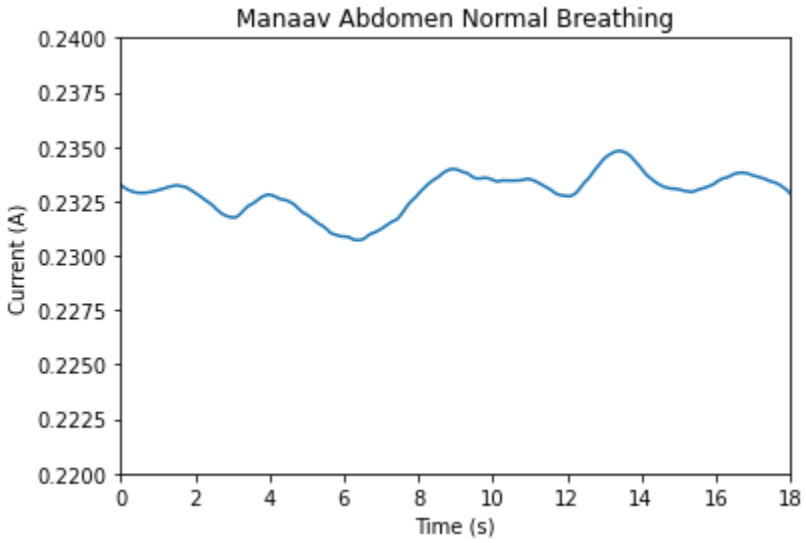


Figure 2. Respiratory Volume and Flow Measurements Courtesy of Costanzo, Physiology 2018.

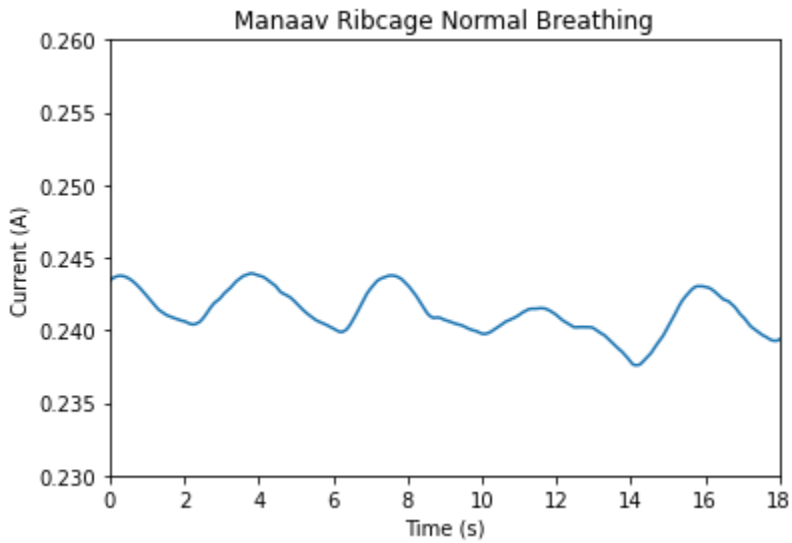
Procedures	1. Weigh out 2g of PDMS base and 0.1g of curing agent in a petri dish. 2. Use a plastic stirring rod to mix the components thoroughly. 3. Weigh out 4g of micromagnet particles of diameter 25 μm using a weigh boat. 4. Mix the micromagnet particle into the PDMS mixture in the petri dish using the plastic stirring rod. Stir the mixture for 10 minutes total. 5. Cure the mixture at 70° C for 4 hours. 6. Peel the cured magnetoelastic layer out of the petri dish. Make sure you are using gloves! 7. Cut out a 2 cm × 2 cm square from the magnetoelastic layer. 8. Tightly fix the magnetoelastic layer on the sample mount of the magnetizer with tape. 9. Magnetize the magnetoelastic layer by an impulse field induced by 350 V voltage. 10. Use a digital Gauss meter to measure the magnetic flux density of the magnetized magnetoelastic layer. 11. Wrap copper wire 10 times around a cylinder with a face big enough to fit the magnetoelastic square. 12. Leave 10 cm of wire before starting the turns. After 10 turns, leave 10 cm of wire at the end. 13. Slide the coil off the cylinder and use tape to keep its shape. 14. Carefully scrape off 1cm of coating from the two wire ends of the conductive coil. 15. Use the Multimeter to measure the resistance of the conductive coil. 16. Stack the magnetized magnetoelastic layer and the conductive coil together. Use tape to attach the two components of the MEG. 17. Repeat steps 1-15 to fabricate an additional MEG. 18. Connect the two MEGs in series by wrapping one wire from each MEG together in aluminum foil. 19. Lay the now connected MEGs on to the middle of the physiotherapy band. Adhere them to the band using tape. Ensure that the two free wire ends hang over the edge of the band. 20. Tie the band tightly to the individual of interest's chest with the MEGs facing inwards. Ensure that the MEGs lie over the diaphragm. 21. Connect the wires of the MEG to the Stanford low-noise current pre-amplifier. 22. Monitor rapid breathing, normal breathing, and forced vital capacity. Record the generated current.

	23. Repeat steps 19-21, with the device shifted upwards so that the MEGs lie across the upper chest. The measurements that we aimed to obtain were tidal volume (TD) from normal/rapid breathing and forced expiratory volume after 1 second (FEV1) and forced vital capacity (FVC) from a hard exhale. Tidal volume is the amount of air that moves in and out of the lungs with each breathing cycle; the average adult male TD is 500 mL and average adult female TD is 400 mL[1]. Forced vital capacity is the amount of air that a subject can forcefully and fully exhale from their lungs[2]. We conducted tests with the MEG device located in two different locations: on the diaphragm and the chest. The diaphragm was chosen due to its role as the main respiratory muscle. The chest was chosen because of its consequential movement from the inflation and deflation of the lungs.
Results	Here we demonstrate a proof of concept of a MEG-based respiratory strain sensor through measurement of the current generated from an MEG placed on the left upper chest (termed ribcage) and an MEG placed immediately below the sternum in the upper middle quadrant of the abdomen (termed abdomen). **A.** 

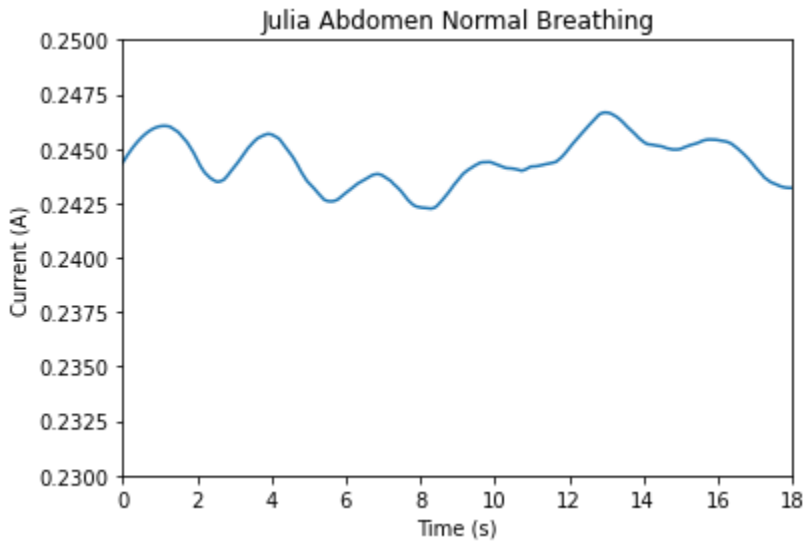
[1] Hallett, Sasha. "Physiology, Tidal Volume." *StatPearls [Internet].*, U.S. National Library of Medicine, 9 May 2021, www.ncbi.nlm.nih.gov/books/NBK482502/.

[2] Healthwise Staff. "Forced Expiratory Volume and Forced Vital Capacity." *Forced Expiratory Volume and Forced Vital Capacity | Michigan Medicine*, www.uofmhealth.org/health-library/aa73564#:~:text=Forced%20vital%20capacity%20(FVC)%20is,that%20are%20measured%20during%20spirometry.

B.


Manaav Ribcage Normal Breathing
Current (A)
0.260
0.255
0.250
0.245
0.240
0.235
0.230
0
2
4
6
8
10
12
14
16
18
Time (s)

C.


Julia Abdomen Normal Breathing
Current (A)
0.2500
0.2475
0.2450
0.2425
0.2400
0.2375
0.2350
0.2325
0.2300
0
2
4
6
8
10
12
14
16
18
Time (s)

D.

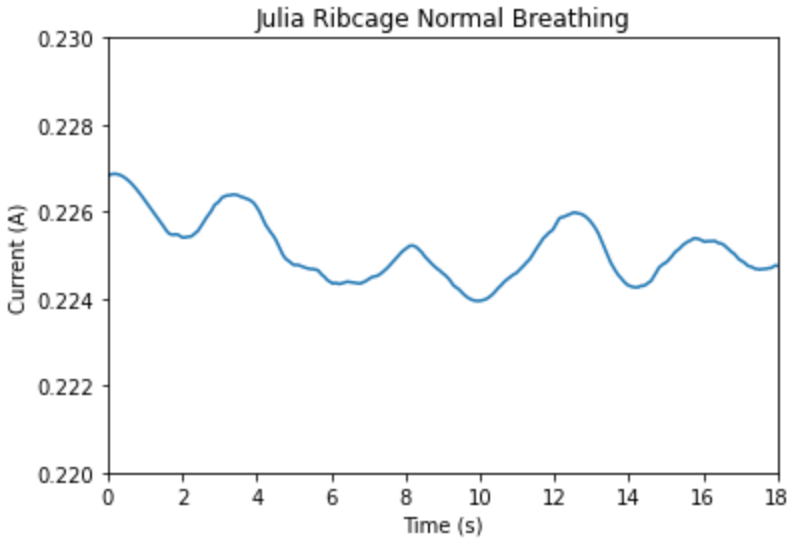
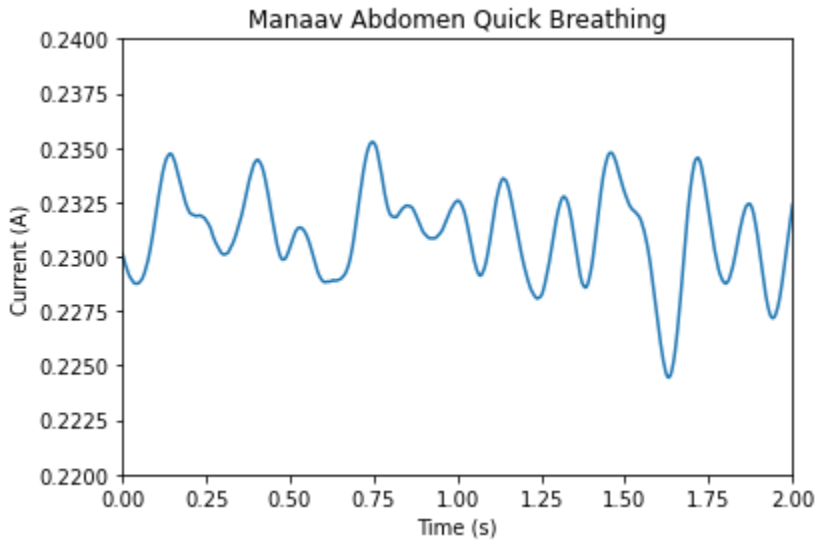


Figure 3. Measurement of Tidal Volume and Normal Breathing Pattern

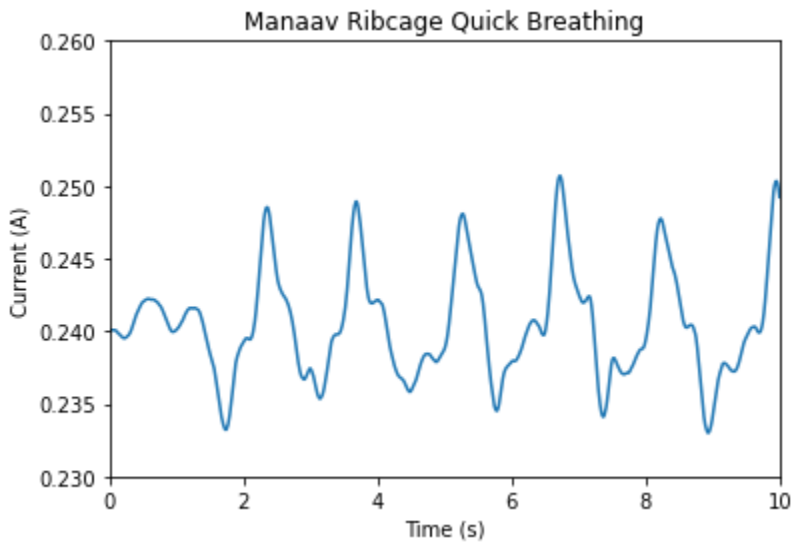
As shown in Figure 3, the average current amplitude of normal breathing for both the ribcage and the abdomen of both participants is approximately 0.005 A. This value remains consistent throughout our trials (including both participants and both locations) indicating that our MEG-based respiratory strain sensor has sufficient sensitivity to detect normal breathing volumes, a value crucial in the long term monitoring of patient physiological status. However, it is important to note that the positioning and how tightly the physical therapy band was wrapped around the participant played a role in the current outputs of the MEGs.

It is important to note that the signal generated by the MEG placed on the rib cage (Figure 3B and 3D) generates a more sinusoidal signal than the MEG placed on the abdomen (Figure 3A and C). This is expected as the muscles associated with the widening of the ribcage, the diaphragm and both sets of intercostal muscles, and involved in both inspiration and expiration while the muscles associated with extension of the abdomen, the abdominals, primarily play a role in forced expiration. Further optimizing the positioning and sensitivity of these two sensor locations can help further deduce information about patient breathing patterns such as the frequency of labored or forced breathing to better diagnose patient health conditions.

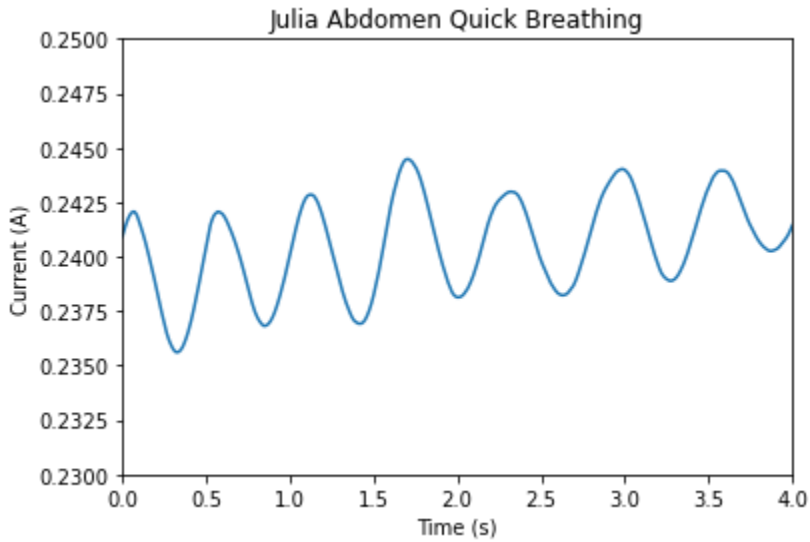
A.


Manaav Abdomen Quick Breathing
Current (A)
0.2400
0.2375
0.2350
0.2325
0.2300
0.2275
0.2250
0.2225
0.2200
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Time (s)

B.


Manaav Ribcage Quick Breathing
Current (A)
0.260
0.255
0.250
0.245
0.240
0.235
0.230
0
2
4
6
8
10
Time (s)

C.



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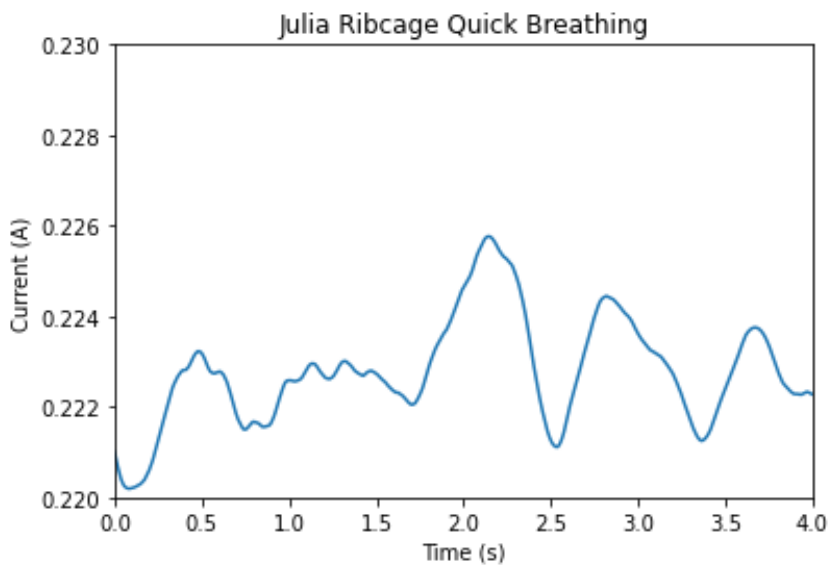

Figure 4. Measurement of Quick (Forced) Breathing and Forced Volume

As shown in Figure 4, the average current amplitude of quick breathing varies depending on the MEG placement and the participant. An MEG placed on Manaav's abdomen (Figure 4A) has an amplitude of approximately 0.005 A, Manaav's ribcage (Figure 4B) has an amplitude of approximately 0.015 A, Julia's abdomen (Figure 4C) has an amplitude of approximately 0.01 A, and Julia's Ribcage (Figure 4D) has an amplitude of approximately 0.006 A. These values vary

drastically depending on the individual patient's proportions which indicates the need for individual strain sensor calibration in the future.

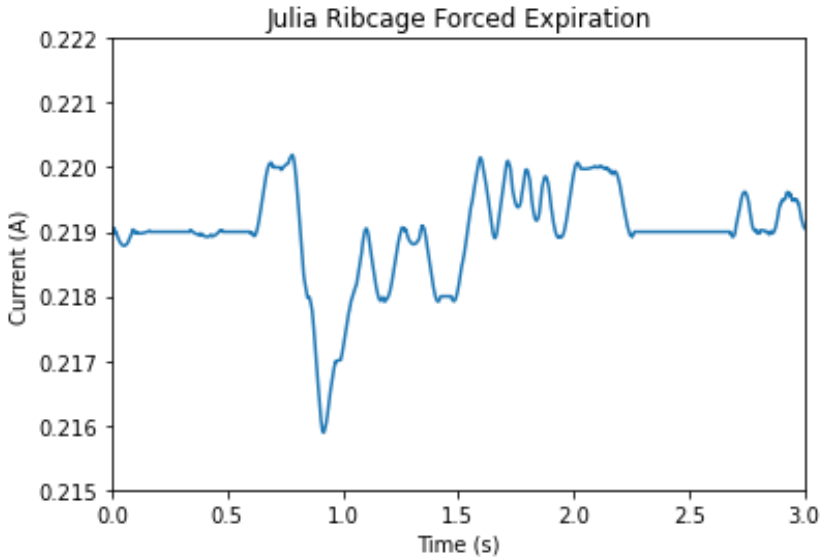

Figure 5. Measurement of Forced Expiration from the Ribcage

Figure 5 displays the current output of a MEG sensor placed on the ribcage during a single forced expiration. The single large drop in current at approximately 0.9 seconds is associated with the diaphragm relaxing causing air to flow out of the lungs. The following signal from 1 second to 2 seconds is associated with the contraction and relaxation of the intercostals and abdominals causing shifts in the position and diameter of the ribcage. Optimizing this sensor placement can improve the forced expiration readings potentially allowing patients to collect pertinent respiratory data at home.

Conclusion

As shown in the experiments we have conducted, the MEG is an optimal solution to the development of a self-powered pleural strain sensor capable of determining the presence of COPD. Due to its advantageous properties including intrinsic waterproofness and ability to directly convert biomechanical motion into electricity the MEG provides a more complete solution to the early diagnosis and data gathering dilemma associated with COPD. However, there are some factors relating specifically to our experimental protocol which could be markedly improved.

While recording the breathing of two subjects, several limitations affected the functioning of the device. Firstly, the physiotherapy bands

	had to be wrapped extremely tight around the subject for the MEG to produce any current, otherwise the motions of the diaphragm and chest were not large enough. Secondly, targeted breathing was required for the MEG to create an electrical signal. For example, when the MEG was connected to the subjects' diaphragm or chest, the subjects had to consciously breathe 'into' their diaphragm or chest. Finally, no signals were created for shallow to normal breathing. As such, the subjects had to breathe either deeply or rapidly for the MEG to be compressed with enough force to create a signal. The hard exhale was easily detected due to the increased acceleration of the muscles of interest which ultimately created a larger force of compression on the MEG. For future developments, it is within our interests to experiment with variations in the formation of the MEG, perhaps altering factors such as thickness, elasticity, size, and shape to optimize the device for more accurate data collection.
References	Chen J. Review Slides. Nov 2021. Chen G, Zhao X, Andalib S, et al. Discovering giant magnetoelasticity in soft matter for electronic textiles. *Matter*. 2021;4(11):3725-3740. doi:10.1016/j.matt.2021.09.012 Chu, M., Nguyen, T., Pandey, V. et al. Respiration rate and volume measurements using wearable strain sensors. npj Digital Med 2, 8 2019. https://doi.org/10.1038/s41746-019-0083-3 Costanzo, Linda S. Physiology. Respiratory Physiology, 2018. Print. Han et al. UpToDate. Chronic obstructive pulmonary disease: Definition, clinical manifestations, diagnosis, and staging, 2021. Print. Zhou Y, Zhao X, Xu J, et al. Giant magnetoelastic effect in soft systems for bioelectronics. *Nat Mater*. Published online September 30, 2021:1-7. doi:10.1038/s41563-021-01093-1